



KOEL FRESH PVT LMT.

MADHYA PRADESH

THE HEART OF INDIAN HORTICULTURE

Winter Internship Report
By Pallati Srivalli

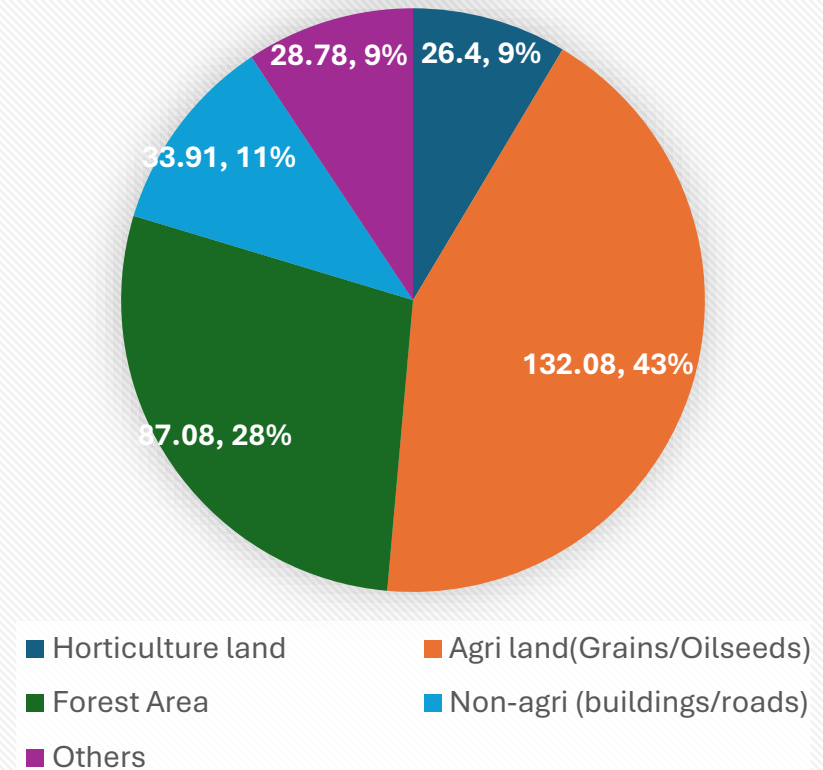


AGENDA : Analysis of the horticulture sector in Madhya Pradesh, emphasizing post-harvest systems and value chains to replicate the Rourkela model

THE WHY?

- National scale opportunity
- Approx. 348 lakh MT Horticulture production
- Largest producer of tomatoes and green peas
- Connectivity to major metro markets
- Mumbai-Delhi connectivity
- 2026 as agriculture welfare year

Land utilisation





Overview

- One of the largest producers of horticultural products
- First in Tomato, Second in Green peas and third largest producer of guava nation-wide
- 11 Agro - climatic zones - Sub-tropical climate with diverse soil types
- High-respiration crops suffer **15%–25%** post-harvest losses without immediate cooling
- 90% of existing cold storage is outdated and **Potato-specific**, leaving high-value fruits and vegetables underserved.
- High Demand due to the increasing losses
- Connectivity to major markets
- A large number of FPOs and SHGs are available to expand our Rourkela model in Madhya pradesh.

TOP 20 CITIES

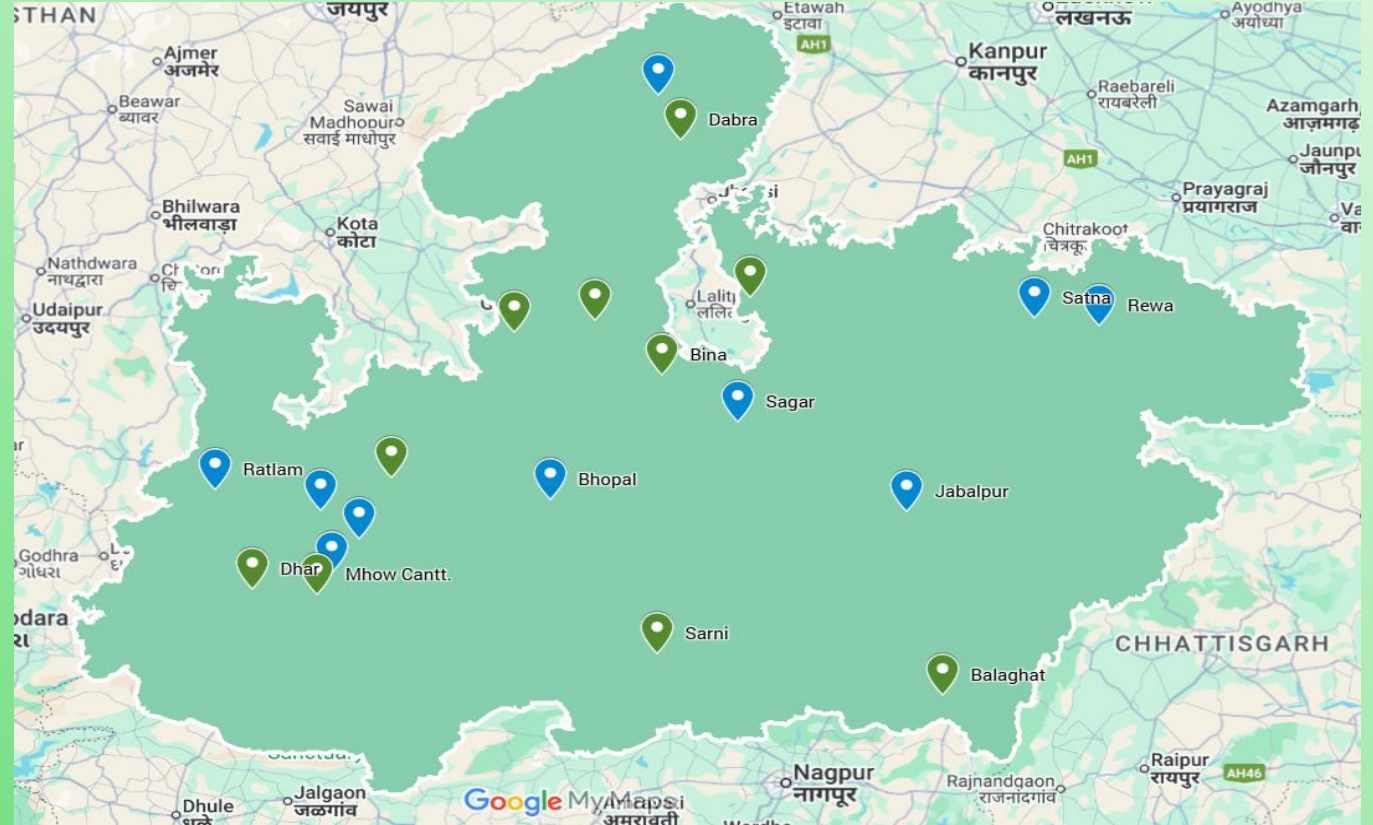
Primary research cities were identified through a weighted evaluation of tier classification, population density, and their role in horticultural trade.

TIER 1

- Indore
- Bhopal
- Jabalpur
- Gwalior
- Ujjain
- Dewas
- Satna
- Sagar
- Ratlam
- Rewa

TIER 2

- Dhar
- Sarni
- Balaghat
- Ashoknagar
- Mhow Cantt.
- Tikamgarh
- Shajapur
- Bina-Etawa
- Raghogarh-Vijaypur
- Dabra



<https://www.google.com/maps/d/edit?mid=1x1DRFa3r4q11ss1IXFQB-1RZfgcZh3A&usp=sharing>



Week-wise Data Collection

- Week 1 – Connectivity Framework
- **Transit Corridor Mapping:**
Assessed road, rail, and air connectivity across 20 cities to identify high-efficiency routes enabling a seamless cold-chain network.
- **Strategic Catchment Profiling:**
Mapped city-level procurement catchments to ensure optimal proximity of units to farm-gate aggregation points.
- **Logistics Bottleneck Analysis:**
Analyzed highway access and transit speeds to reduce last-mile delays and mitigate post-harvest crop stress.

- Week 2 – Institutional Framework
- **Regulatory Compliance Audit:**
Reviewed municipal corporations and development authorities to ensure site selection aligned with local zoning and land-use regulations.
- **APMC Institutional Directory:**
Developed a comprehensive database of primary mandis and mandi officials to enable direct institutional engagement and market integration.
- **Policy & Governance Mapping:**
Mapped key administrative stakeholders to facilitate future approvals and support sustained regulatory compliance.



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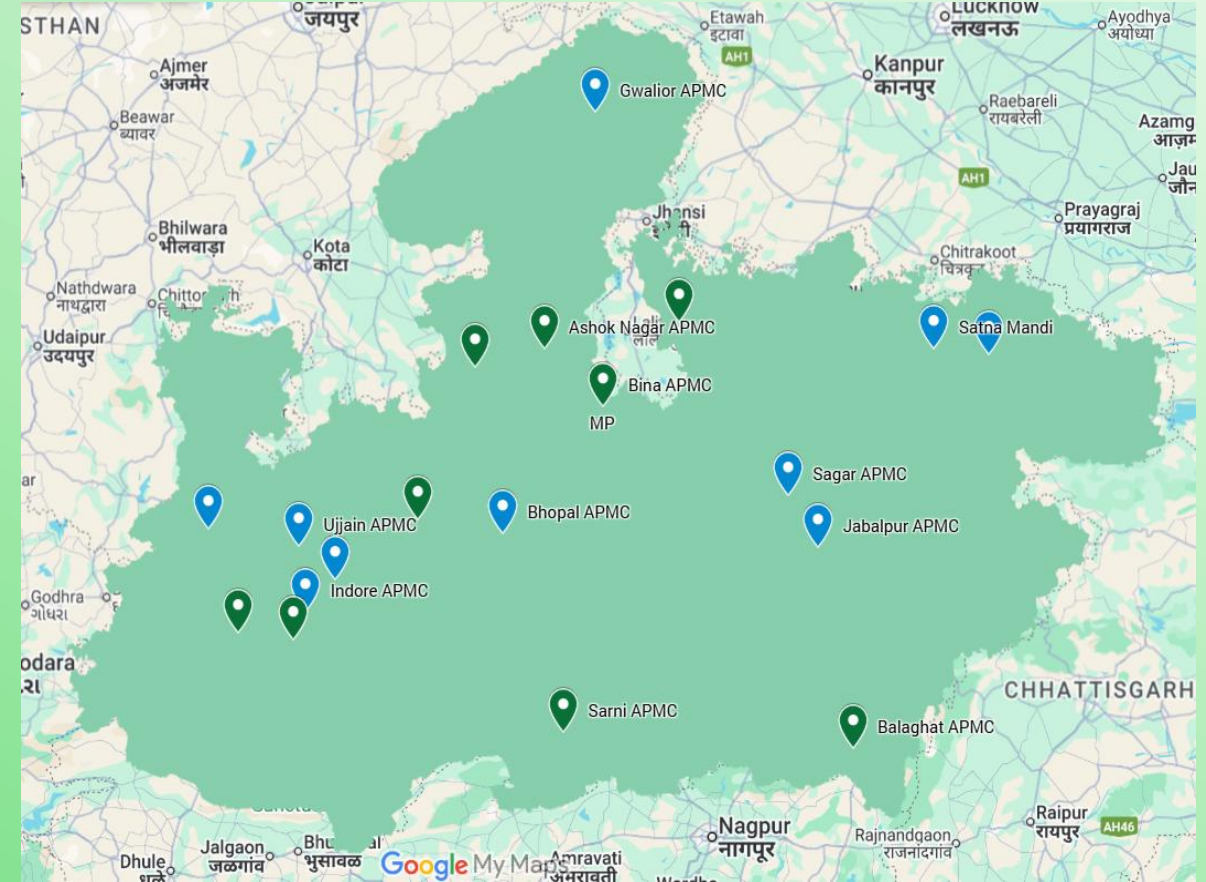
- Week 3 - Policy, Finance & Infrastructure Inventory
 - **Government Scheme Audit:** Evaluated AIF, MIDH, and PMFME schemes to leverage financial subsidies and maximize efficiency.
 - **Fiscal Incentive Mapping:** Audited state-specific agricultural grants to align project funding with available government fiscal support.
 - **Existing Infrastructure Inventory:** Documented approved cold chain projects to identify underserved regions and avoid market over-saturation.
- Week 4 – Trend Analysis & Stakeholder mapping
 - **Loss & Trend Analysis:** Analyzed 3-year production data and commodity wastage to prioritize locations with the highest need for cooling intervention.
 - **Institutional Directory:** Compiled contact lists for Horticulture Officers and APMC Boards to facilitate grassroots technical and policy coordination.
 - **Aggregation Partner Mapping:** Cataloged NRLM SHGs and FPOs to identify community-led operational partners for decentralized unit management.

Madhya Pradesh APMC Network & Strategic Coverage

Division-wise, district-wise, category-wise agricultural produce market committees of the state

As on date (04.01.2024)

Name of the division	Number of districts	Category wise (Category) Agricultural Produce Market Committees Number					Total Subdivisions
		"C" category	"B" category	"C" category	"D" category	Total Mandis	
Bhopal	8	12	11	8	18	49	57
Indore	8	6	8	8	12	34	62
Ujjain	7	10	5	8	19	42	50
Gwalior	8	3	6	11	25	45	39
Sea	6	3	4	10	19	36	21
Jabalpur	9	4	7	7	17	35	47
Reva	9	1	1	4	12	18	22
sum	55	39	42	56	122	259	298



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Satna (The Tomato Scaling Engine)

Strategic Scale-Up of the Rourkela Tomato Blueprint

•**Production Anchor:**

Utilizes Satna's ODOP (One District One Product) status for tomatoes to ensure a concentrated, high-volume supply for decentralized units.

•**Reducing Post-Harvest Losses:**

Addresses the 15%–25% wastage at Sabji Mandi through rapid pre-cooling, stabilizing high-respiration crops.

•**Optimized Logistics & Connectivity:**

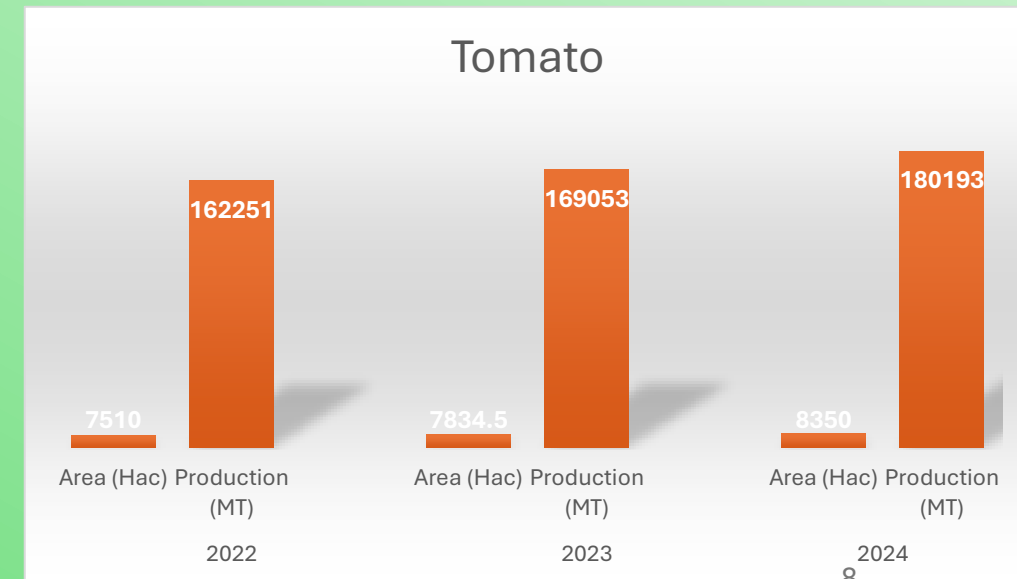
Leverages NH-30 (Varanasi–Jabalpur) and Satna Railway Junction for cost-efficient bulk transport to North Indian metros.

•**FPO & SHG Integration:**

Collaborates with Sohawal Krishak Producer Co. and local SHGs to establish a resilient hub-and-spoke collection network at the farm-gate.

•**Model Replication:**

Scales the Rourkela “Cooling-as-a-Service” pilot to a regional and national commercial model.





Rewa (The High-Value Fruit Frontier)

Premium Value-Addition & Regional Export Hub

- Production Anchor:**

Captures the regional premium guava and mango market, shifting from basic preservation to high-margin fruit grading.

- Quality Optimization:**

Reduces 30% bruising and browning losses, maintaining export-grade quality for high-end retailers and processors.

- Efficient Logistics & Connectivity:**

Leverages Chorahata Transport Nagar and Rewa–Satna Bypass to integrate with national refrigerated trucking networks.

- FPO & SHG Collaboration:**

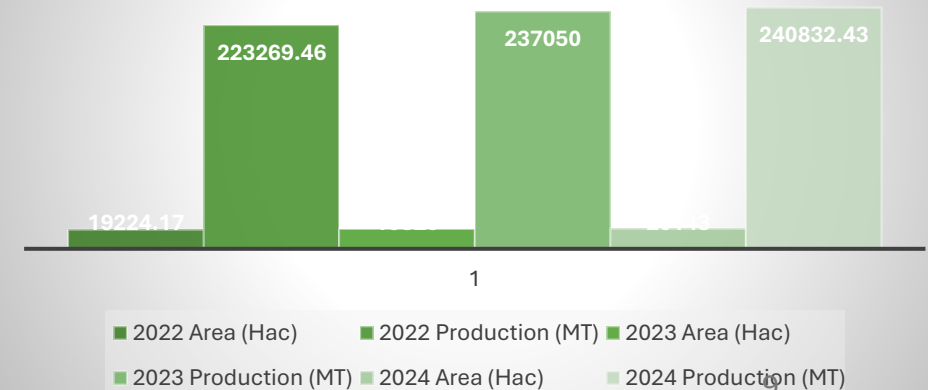
Engages Rewa Block SHGs for primary aggregation and partners with regional horticulture FPOs for large-scale B2B market linkages.

- Model Evolution:**

Transforms the Rourkela approach from “Survival Cooling” to “Market-Linkage Cooling,” boosting profit per crate for community partners.



Total Fruits





Most Suitable cities

Jabalpur – The National Specialty Hub

- **Production & Focus:** Premier cluster for Green Peas & high-grade Cabbage; dominates national harvest window.
- **Wastage Gap:** High-respiration crops lose 20%+ value without cooling.
- **Logistics & Connectivity:** NH-7 enables direct refrigerated transit to Hyderabad & Bangalore.
- **Rourkela Model Fit:** Replicates “Specialty Crop” approach using dense NRLM SHG networks for fast village collection.

Indore – The Commercial Distribution Gateway

- **Production & Focus:** State’s largest hub for Leafy Greens & Potato; links Malwa production to organized retail.
- **Wastage Gap:** High urban transit temps cause rapid wilting; decentralized cooling extends freshness 3–5 days.
- **Logistics & Connectivity:** Indore Air Cargo & Pithampur SEZ support interstate/export-grade perishables.
- **Rourkela Model Fit:** Evolves “Farm-gate Survival” to “Last-Mile Distribution” via FPOs, bypassing middlemen.

Ujjain – The Aggregation & Corridor Anchor

- **Production & Focus:** High-velocity Indore–Ujjain corridor; aggregates Potato, Onion & Tomato for Western India.
- **Wastage Gap:** Cold stores at 90%+ capacity; small farmers face 15% losses.
- **Logistics & Connectivity:** Near Western Railway & Vikram Udyogpuri for bulk transport to Gujarat & Maharashtra.
- **Rourkela Model Fit:** Provides “Overflow Cooling” during peak gluts via decentralized tech & Malwa FPOs.

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Dewas – The Malwa Production Powerhouse

- **Production & Focus:** High-density satellite hub for Potato, Onion, Garlic, and seasonal Gourds serving the Indore urban corridor.
- **Wastage Gap:** 20–25% losses at harvest peaks, driven by lack of farm-gate pre-cooling and field-heat removal before Indore Mandi.
- **Logistics & Connectivity:** Strategic junction of NH-3 (Agra–Mumbai) and Dewas–Bhopal Highway, linking two major consumer markets.
- **Rourkela Model Fit:** Acts as a short-term buffer managed by Dewas Block SHGs, enabling 48–72 hour storage to bypass mandi price crashes.

Ratlam – The Delhi–Mumbai Logistics Pivot

- **Production & Focus:** ODOP district for Garlic; key exporter of Grapes and Mandarin Oranges.
- **Wastage Gap:** Highest loss at ~30%, driven by absence of humidity-controlled Garlic storage and 0°C–2°C cold chain for Grapes.
- **Logistics & Connectivity:** Critical node on the Delhi–Mumbai Expressway with a major rail junction, cutting transit time to ports and NCR by ~50%.
- **Rourkela Model Fit:** Evolves into a quality export center with decentralized pre-cooling and primary processing for Ratlam Garlic FPOs.

5 pitch points for the expansion in Madhya Pradesh

- Systemic Wastage Mitigation:**

Decentralized multi-commodity pre-cooling combined with post-harvest handling awareness directly addresses the state's 25% wastage caused by storage deficits and field-heat losses, ensuring produce quality, nutritional integrity, and operational continuity.

- ODOP-Aligned Policy Acceleration:**

Locating cooling infrastructure at ODOP hubs such as Satna (Tomato) and Ratlam (Garlic) preserves export-grade quality at source, converts policy intent into higher farmer incomes, and enables access to structured subsidy mechanisms.

- NRLM/SHG-Led Social Transformation:**

Operating "Cooling-as-a-Service" units through NRLM-backed SHGs establishes women-led technical entrepreneurship while ensuring trusted, localized management of cold-chain assets.

- Logistics Optimization & Transit Velocity:**

Positioning units as pre-cooling gateways along expressway and rail corridors enables faster market access, improves price realization, and integrates regional supply into the national high-speed food network.

- Digital Governance & Market Intelligence:**

IoT-enabled, solar-hybrid systems generate real-time harvest and quality data that support informed market decisions, transparent price discovery, and data-driven expansion planning.



Conclusion

- Madhya Pradesh – The Optimal Growth Frontier:**

MP is a top national producer of perishable horticulture crops. Combined with extreme seasonal temperatures and limited farm-gate cooling, it offers a high-demand market with enormous potential for post-harvest loss reduction.

- Strategic Expansion: Satna & Rewa:**

Launching in the Satna–Rewa cluster leverages ODOP status and high-density production of Tomato and Guava. This enables seamless replication of the Rourkela community-managed model, delivering higher commercial margins while maximizing government subsidy support.



THANK YOU